

DescSpim Deepsky Construction manual

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1. Microscope Construction

All below mentioned parts can be purchased at Thorlabs unless otherwise stated.

Assembling the detection arm stage components

1. Obtain a crossed-roller side-actuated translation stages (XR25P/M), loosen the flexure lock, and remove the micrometer head.
2. If necessary, convert the stage to a left-hand configuration by swapping the side mounts with the locking plate.

CRITICAL STEP Fully retract the micrometer head before removing the locking plate to prevent the stage from rapidly retracting.

3. Mount one motorized actuator (Z825B) onto the translation stage, labeling the assembly as the camera stage
4. Fix the stage to the optical breadboard at the given location (Fig x) with M6 cap screws
5. Attach a 66 mm dovetail rail (XT66SD-250) to the translation stage with M6 cap screws.
6. Mount 3 rail carriages (XT66P2/M) on the 66 mm dovetail rail.
7. Fix three post holders (PH30/M) to the rail carriages with M6 setscrews

Assembling the detection objective and filter slider

8. Attach the W26 x 0.706 internal to SM1 external thread adapter (SM1A27) to the detection objective (MY10X-803).
9. Attach the objective thread adaptor to the SM1 internal to SM2 external thread adapter (SM2A6).
10. Attach the SM2 external thread adaptor to the 60 mm cage plate, 0.5" thick (LCP34/M).
11. Insert two 2" long cage assembly rods (ER2) in the bottom two holes of the 60 mm cage plate and tighten the M4 set screws.
12. Attach a 40 mm optical post (TR40/M) to the cage plate using an M4 set screw.
13. Attach the SM1-threaded 30 mm to 60 mm cage plate adapter, 0.5" thick (LCP33/M) via the bottom two holes to the cage assembly rods and tighten the M4 set screws.
14. Insert four 1" long cage assembly rods (ER1) into the centre four holes of the SM1-threaded 30 mm to 60 mm cage plate adapter.
15. Insert the filter slider (ELL9) between the 60 mm cage plate and SM1-threaded 30 mm to 60 mm cage plate adapter, fastening to the four previously inserted 1" long cage assembly rods.

16. Mount the necessary chromatic filters in SM1 lens tubes (SM1L03) and thread into the filter slider.
17. Attach a 40 mm optical post (TR40/M) to the SM1-threaded 30 mm to 60 mm cage plate adapter with an M4 set screw.

Assembling the tube lens and camera

18. In order, screw together the following components: 30 mm cage plate (CP33/M), SM1 external to SM2 internal thread adapter (SM1A2), 0.5" SM2 lens tube (SM2L05), $f = 200$ mm tube lens (TTL200-A), 3" SM2 lens tube (SM2L30), 0.5" SM2 lens tube (SM2L05), SM1 external to SM2 internal thread adapter (SM1A2), 30 mm cage plate (CP33/M).
19. Insert four 1" cage assembly rods into the second 30 mm cage plate and tighten the retaining screws.
20. Fasten a second 30 mm cage plate onto the cage assembly rods, leaving a small gap between the two.
21. Screw a SM1 coupler (SM1T10) onto the second 30 mm cage plate.
22. Screw a 0.5" SM1 lens tube (SM1L03) onto the SM1 coupler.
23. Screw the Kiralux 8.9 MP Monochrome CMOS camera (CS895MU) onto the 0.5" SM1 lens tube.
24. Retaining ring.

CRITICAL STEP The assembly allows for the camera to be rotated to the vertical position to maximise the field-of-view.

CRITICAL STEP The precise alignment of the camera and tube lens is critical for diffraction limited imaging. The distance between the camera and tube lens can be adjusted by loosening the retaining screws on the 30 mm cage plate closest to the camera and moving the camera attached to the cage plate along the optical path direction. To ensure optimal alignment, point the camera outside the window and align the camera by focusing on structures 50 meters away.

Mounting the detection arm to the stage

25. Attach the detection objective and filter slider to the camera and tube lens assembly using the four 1" long cage assembly rods and tighten the retaining screws.
26. The complete detection arm can now be attached to the detection arm stage using the three 40 mm optical posts and post holders.

CRITICAL STEP Ensure the detection arm is level before fixing in place using a circular bullseye level (LVL03).

Assembling the sample stage

27. Fasten a 3" dovetail rail (XR25DR2) flush with the corner of the optical breadboard using M6 set screws.
28. Obtain a crossed-roller side-actuated translation stages (XR25P/M), loosen the flexure lock, and remove the micrometer head.
29. If necessary, convert the stage to a left-hand configuration by swapping the side mounts with the locking plate.

CRITICAL STEP Fully retract the micrometer head before removing the locking plate to prevent the stage from rapidly retracting.

30. Mount one motorized actuator (Z825B) onto the translation stage, labeling the assembly as the sample stage.
31. Mount the translation stage onto the dovetail rail.
32. Fix two 25 mm pillar posts (RS25/M) to the back two counterbored M6 holes on the underside of the Self-Contained XYZ 25 mm Translation stage (LX30/M) using M6 cap screws.
33. Fix two 25 mm pillar posts (RS25/M) to the front two M6 counterbored holes of the translation stage using M6 set screws.
34. Fasten the Self-Contained XYZ 25 mm Translation stage to the two posts on the front of the translation stage using M6 cap screws.
35. Create a ledge for the cuvette using a mounting base BA1/M attached to the vertical translational stage using an M6 cap screw.
36. Fasten a microscope slide spring clip to the vertical translation stage using an internal M4 x 0.7 threads and external M6 x 1.0 threads adapter (AE4M6MS).

Assembling the illumination arm stage components:

37. Obtain two crossed-roller side-actuated translation stages (XR25P/M). From one translation stage, loosen the flexure lock, and remove the micrometer head.
38. If necessary, convert the stage to a right-hand configuration by swapping the side mounts with the locking plate.

CRITICAL STEP Fully retract the micrometer head before removing the locking plate to prevent the stage from rapidly retracting.

39. Fix the stages to the optical breadboard at the given location (Fig x) with M6 cap screws.
40. With the micrometer fully retracted fasten a 500 mm long, 66 mm wide single dovetail rail (XT66SD-500) to the translation stages with four M6 cap screws.
41. Mount 4 rail carriages (XT66P2/M) on the 66 mm dovetail rail.

42. For the illumination cage, fix three post holders (PH30/M) to the front three rail carriages with M6 setscrews.
43. For the galvo mirror, fix one $\varnothing 12.7$ mm pedestal post holder (PH30E/M) to the rear rail carriage with a clamping fork (CF038-P5) and M6 cap screw.

Fiber and cylindrical lens assembly

44. Insert three cage rods (4" length, ER4) into an XY adjustable lens-mount (CXY1A), tighten the retaining screws and insert 50 mm collimating lens (AC254-050-A).
45. Insert the SM1FC2 fiber mount with the single mode fiber from the laser combiner attached into a 30 mm cage plate (CP33/M) and slide on to the cage system so that the flat side of the collimating lens is facing the fiber.
46. Collimate the laser beam by moving the fiber mount along the optical axis and use a shear plate to verify.
47. Roughly XY-align the beam exiting the collimating lens by adjusting the setting screws on the lens holder.
48. Slide a 30 mm cage-compatible adjustable pinhole (CP20D) flush against the lens mount. This pinhole can be used for alignment purposes (See CRITICAL STEP) and should be fully open otherwise.
49. Slide a 30 mm cage system adjustable slit (VA100CP/M) onto the cage rods with the slit orientated parallel to the optical path (see next steps).
50. Attach the cage system to a right-angle kinematic elliptical mirror mount (KCB1EC/M) with mirror (BBE1-E02) inserted and tighten the retaining screws.
51. Insert four cage rods (4" length, ER4) on the other side of the mirror mount and slide two 30 mm cage plate (CP33/M) in them with the SM2 thread facing down.
52. Attach another right-angle kinematic elliptical mirror mount with mirror inserted onto the end of the cage rods.
53. Insert three cage rods (3" length, ER3) on the other side of the mirror mount facing down and slide a cage-compatible rotation mount (CRM1PT/M) with the threads facing away from the mirror mount.

CRITICAL STEP The collimated laser beam coming from the two mirrors needs to be in the center of the rotation mount to ensure optimal alignment through the subsequent cylindrical lens. To align, close the alignment pinhole after the collimation lens and walk the beam using the two mirrors through a frosted alignment disk (DG10-1500-H1-MD) threaded into the rotation mount.

54. Insert a 50 mm cylindrical lens (ACY254-050-A) into an SM1 lens tube (SM1L10). It will be inserted into the rotation mount during the alignment procedure.

Illumination cage assembly

55. Assemble three cage rods to the length of 500 mm and insert into the following cage compatible components in the following order: two cage plates (CP33/M), an XY-adjustable lens mount (CXY1A), two cage plates (CP33/M), and another XY-adjustable lens mount. The last lens mount constitutes the front of the assembly and will be used later for mounting the illumination objective lens. It should roughly be located at the end of the cage rods.
56. From the back of the assembly, slide the first cage plate to the end of the rods.
57. Position the next cage plate at approximately 60 mm from the end of the rods.
58. Position the center lens mount approximately 170 mm from the end of the rods.
59. Position the next two cage plates at approximately 290 mm and 390 mm from the end of the rods respectively (opposite of the front where the objective mount is).
60. Slide an 8" cage rod (ER8) into the system from the back to connect the back of the cage and the first lens mount.
61. Slide another 10" cage rod (ER10) from the front into the system.

CRITICAL STEP The resulting gap allows to mount the lens into the middle lens mount at a later time to improve alignment efficiency.

62. Attach three 40 mm posts (TR40/M) to all the 30 mm cage plates except the plate at the back of the assembly and mount the assembly on the front three post holders on the dovetail rail.
63. Attach two 75 mm posts (TR75/M) to the cage plate threads on the fiber + cylindrical lens assembly using an M4-M6 adapter and mount two Snap on 30 mm cage mounting brackets (CPMA3) on the bottom (CP33B cage mounting brackets can also be used, see "Aligning the illumination components").
64. Mount the galvo mirror (GVS211/M) on a 30 mm post (TR30) and insert into pedestal post at the back of the dovetail rail.

Connecting the electrical components

65. Route each stage's motor cable to its KDC101 controller, plug each controller into its KSP101 power supply, and arrange the controller–power pairs neatly on the breadboard.
66. Test the connection and function of the KDC101 controller using Thorlabs Kinesis software and make sure that both sample and camera stage are movable.
67. Test the ELL9 filter slider by plugging in the USB cable and start ELLO software and make sure slider can move into all positions without problem.

CRITICAL STEP If the slider seems to move slowly through the positions or gets stuck it is possible to apply a very small amount of lubricant (WD-40) on the rail where the slider is carried. However, no lubricant should be reaching the bottom black part where the piezo elements grab the slider.

68. Connect the single mode fiber to the fiber coupler of the laser system and test using the correct software (here, Cobolt monitor was used).

CRITICAL STEP Lower the laser power for alignment purposes to a minimum (~1-5 mW).

69. Connect the camera to the computer and test using the software Thorcam.
70. Connect the galvo mirror to its power supply
71. Set the JP7 Volts/Degree scaling factor control pin to 0.8 V/° (see pin diagram [here](#)).
Test by supplying 2V and 200 Hz and viewing the beam profile.

Aligning the illumination components

72. Close the circular pinhole underneath the fiber collimating lens to a minimum.
73. Open the slit to the maximum.
74. The laser beam should enter the rotation mount through the center assuming correct beam steering with the two elliptical mirrors (See CRITICAL STEP “Fiber and cylindrical lens assembly”)
75. Ensure that the fiber and cylindrical lens assembly is mounted central on the lower illumination cage by mounting the two posts central on the Snap-on mounting brackets. The mounting bracket CP33B offers a central bore hole and seems to be better suited, however, it was released before the writing of this manual but after designing the system.
76. Adjust the setting screws on both adjustable lens holders to the center position.
77. Mount a frosted alignment disk (DG10-1500-H1-MD) into the front adjustable lens tube where the objective lens will be later.
78. Position 30 mm Cage alignment plates (CPA1) between the mounting brackets and between the two cage plates in front of the objective lens holder.
79. Attach a slip-on optical post collar (R2) to the post attached to the galvo mirror.
80. Loosen the screw of the galvo pedestal post holder, adjust the height of the galvo mirror, and tighten the screw again. After adjustment, reset the slip-on optical post collar.
81. Adjust first rotation of the galvo by loosening the screw of the galvo pedestal post holder and rotate the post to the desired rotation without adjusting the height (using the slip-on optical post collar) and tighten the screw again.
82. Loosen the screw on the galvo mount, adjust and tighten again.

CRITICAL STEP Iterate steps 78-80 until the beam hits the center of the galvo mirror AND is passing through both alignment plates and the frosted alignment disk.

83. Mount an alignment cross at a far distance from the frosted alignment disk behind the sample stage so that the aligned beam sits in the center of the cross.
84. Insert cylindrical lens into rotation mount by facing the flat side towards the galvo.
85. Rotate the cylindrical lens so that the laser line shown on the galvo is oriented vertically (parallel to the optical axis in space)
86. Tighten/loosen the rotation mount setting screws and adjust z-axis alignment of the cylindrical lens by focusing the line on the galvo mirror surface.
87. Insert 200 mm lens (AC254-200-A) into central adjustable lens mount and adjust the XY-screws until the beam sits in the center of the alignment cross.
88. Mount the illumination objective and align to the center of the alignment cross with the XY-screws of the lens holder.

2. Sample preparation

All incubations should be performed with sufficient solution volume to fully immerse the sample. Gentle agitation during staining and washing steps is recommended to improve reagent exchange. Protect fluorophore-containing solutions and stained samples from light whenever possible.

Adult mouse lung

Here, adult mouse lung tissue was prepared from transgenic B6N.Cg-Tg(Sftpc,-EGFP)1Dobb/J mice and stained prior to CUBIC-X clearing.

Tissue collection and fixation

1. Dissect the adult mouse lung from B6N.Cg-Tg(Sftpc,-EGFP)1Dobb/J mice.
2. Fix the tissue overnight in 4% paraformaldehyde (PFA).
3. Wash the fixed tissue in PBS.

Tissue sampling

4. Using a 3 mm biopsy punch, prepare a 3 mm tissue piece from the fixed lung.

CRITICAL STEP Ensure the tissue piece is uniformly cut to facilitate consistent staining, antibody penetration, and clearing.

Pre-incubation

5. Incubate the tissue sample in staining buffer for 30 min before staining.

Primary fluorescent staining

6. Here we prepared staining buffer supplemented with: Atto594 NHS Ester, 1:100 dilution (ATTO-TEC GmbH), Propidium iodide, 1:500 dilution (Invitrogen)
7. Incubate the sample in this staining solution overnight at 4 °C.

CRITICAL STEP NHS esters are moisture-sensitive and react with primary amines. Prepare the staining solution fresh and minimize unnecessary exposure to aqueous conditions before use.

Post-staining washes

8. Wash the sample 1 × 30 min in staining buffer.
9. Wash the sample 2 × 30 min in PBS.

GFP signal enhancement

10. Prepare HEPES-TSC buffer supplemented with FluoTag®-X4 anti-GFP, 1:200 dilution (NanoTag Biotechnologies).
11. Incubate the sample in this solution overnight at 35 °C.

CRITICAL STEP Maintain the incubation temperature at 35 °C during FluoTag®-X4 anti-GFP labeling to ensure efficient signal enhancement without compromising tissue integrity.

Post-labeling washes

12. Wash the sample 3 × 30 min in labeling buffer.

Clearing

13. Process the stained sample in CUBIC-X1 and CUBIC-X2 as described above.

CRITICAL STEP Ensure that excess staining and labeling reagents are thoroughly removed before clearing, as residual fluorophores or binding reagents may increase background or interfere with optical transparency.

Embryonic kidney

Here, embryonic kidneys were dissected from P17 embryos, stained, and cleared using the CUBIC-X workflow.

Tissue collection and fixation

14. Dissect embryonic kidneys from P17 embryos.
15. Fix the tissues overnight in 4% paraformaldehyde (PFA).
16. Wash the fixed tissues in PBS.

Pre-incubation

17. Prepare staining buffer containing Tetrahydrofuran (THF), 30% ($\geq 99.9\%$, anhydrous, inhibitor-free; Sigma-Aldrich), NaCl, 0.5 M in 0.05 M acetate buffer, pH 5.

18. Pre-incubate the samples in this staining buffer for 30 min.

CRITICAL STEP Use anhydrous, inhibitor-free THF as specified, since solvent quality can affect staining consistency and tissue compatibility.

CRITICAL STEP Confirm that the staining buffer pH is 5.0, as deviations in acidity may alter NHS ester labeling efficiency.

Fluorescent staining

19. Prepare staining buffer supplemented with Alexa Fluor 555 NHS Ester, 1:100 dilution (Thermo Fisher Scientific), Propidium iodide, 1:500 dilution (Invitrogen)

20. Incubate the samples in this staining solution overnight at 4 °C.

CRITICAL STEP Prepare NHS ester-containing staining solutions fresh immediately before use to preserve reactivity.

Post-staining washes

21. Wash the samples 1 × 30 min in staining buffer.

22. Wash the samples 2 × 30 min in PBS.

Clearing

23. Process the stained samples in CUBIC-X1 and CUBIC-X2 as described above.

CRITICAL STEP Incomplete washing after staining may increase background fluorescence during imaging after clearing.

3. Hardware-Software Control

Install MCC

1. Connect all devices to the computer and close all device-individual software (Kinesis, ELLO, Thorcam, etc.).
2. Make sure to note all COM ports of the devices as they will be needed for the next step.
3. Follow the “Mission Control Center: Installation and Operation Manual” to install the MCC on the computer.

Image acquisition

4. Place the sample into the sample stage and roughly find the sample.
5. Start MCC and connect to all devices.
6. Set the camera exposure time/, laser power per laser, movement parameters of the translation stages, filter positions and galvo settings.

Single color imaging

7. Focus on the sample.
8. Use Start (test) or Start (opposite) in the stage control panel to move to the desired starting position, then press Register Start Position to save the start position.
9. In the camera panel, select the Mode, and if you want to use automatic return to the start position, check auto-return.
10. Press the Start Recording button. When the stage reaches the end of the desired imaging range, press Start Recording (Stop) again to stop acquisition. A save dialog will appear; save the data to any location.

Multi-color imaging

11. Confirm that the stages have returned to the start position and have stopped. If they have not returned, press Go to Start Position.
12. Assign the filter to the correct laser by changing the laser wavelength in the laser panel.
13. Check Synchronize camera stage, and when you switch the filter slot, focus adjustment is performed automatically.
14. Check the number of frames acquired in the first track, enable Auto-stop at N frames, and enter the same number.
15. Repeat steps 12-14 to add more tracks.
16. Press the Start Recording button in the camera panel. Acquisition will stop automatically, and a save dialog will appear; save the data to any location.